

RICKY (RUIQI) DING

neuroai-research.github.io/Ricky · github.com/tesla-cat
e0134117@u.nus.edu · linkedin.com/in/rickyding1997

RESEARCH INTERESTS

I am broadly interested in AI and neuroscience/neurotechnology research. My current self-directed focus is neuroscience-inspired AI: understanding how specific modules and circuits of the human brain achieve their functions, and replicating those functions with modern AI methods, in the spirit of [Hassabis et al. \(*Neuron*, 2017\)](#). I trained in physics and have since worked as a software developer.

CURRENT RESEARCH PROJECT

NeuroAI Research

2026 – present

([website](#), [code](#))

- Self-directed study towards replicating brain modules and circuits with modern AI methods; currently in the learning and re-implementation stage
- Minimal re-implementations of models from Dayan & Abbott, *Theoretical Neuroscience*: neural encoding and decoding, information theory, network models, plasticity, reinforcement learning
- Minimal re-implementations of GPT-2, Gemma 2, PPO, SAC, DreamerV3 and flow matching

EDUCATION

National University of Singapore

2016 – 2020

B.Sc. (Honours) in Physics, Highest Distinction

Singapore

- Thesis (supervisor: Prof. Berthold-Georg Englert): *Density Potential Functional Theory in x and p spaces and its implementation*
- Scholarship: [Science & Technology Undergraduate Scholarship](#)
- Degree certificate: [image](#); [OpenCert](#), verifiable at opencerts.io

University of Göttingen

2019

Undergraduate exchange programme (non-degree)

Göttingen, Germany

- Took courses in biophysics from the M.Sc. Physics programme ([certificate](#))

RESEARCH EXPERIENCE

Centre for Quantum Technologies, National University of Singapore

2020 – 2021

Research Assistant (Software Development)

Singapore

- Compared deep reinforcement learning with gradient ascent pulse engineering (GRAPE) for quantum optimal control ([report](#))
- Studied transmon (superconducting) quantum computing ([notes](#))
- Worked on IQ mixer tuning ([notes](#))
- Employment [certificate](#)

Research Internships, National University of Singapore

2018 – 2019

Undergraduate research

Singapore

- 2019: [Quantum Computing Lab](#), supervised by [Prof. Dzmitry Matsukevich](#)
- 2018: Quantum Theory Group, supervised by [Prof. Berthold-Georg Englert](#)

PROFESSIONAL EXPERIENCE

Frontline Industrial Software

2022 – 2025

Software Developer

Singapore

- Developed software products for industrial clients, applying:
 - large language models for content generation
 - vector embeddings and similarity search
 - automatic differentiation and stochastic gradient descent (Adam)
 - density-based clustering ([HDBSCAN](#))
 - mixed-integer linear programming ([PuLP](#))
- Frontend: Vue, React, Capacitor (Web, Android, iOS); backend: Python, JavaScript (FastAPI, Bun)

PERSONAL PROJECTS

Algorithmic Trading

- Differentiable backtesting; deep reinforcement learning

Tesla Coil Gun ([video](#))

- [Physics simulation](#) of a dual-resonant solid-state Tesla coil (DRSSTC); [waveform synthesiser](#)
- [Schematic](#) and [PCB](#) design; CPLD programming in Verilog